THE QUOTIENT FUNCTORS SATISFY THE EQUALIZER PROPERTY OVER FPQC COVERINGS OF X

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Situation situation-q (canonical tag 082M). cited at statement line 2127. Standing situation; fpqc covering of X with the two-step coverings.
- Remark remark-q-sheaf (canonical tag 08IU). cited at proof line 2149. The glued surjection F_T -> Q.
- Lemma spaces-morphisms-lemma-flat-permanence (canonical tag 05VY). cited at proof line 2153. Q flat over T.
- Lemma spaces-descent-lemma-finite-presentation-descends (canonical tag 060V). cited at proof line 2159. Q of finite presentation.

Source reference. Original definition 082M: sources/stacks-chow/tags.txt, source lines 10257–10257. Cited at statement line 2127: standing situation.

Source reference. Original remark 08IU: sources/stacks-chow/tags.txt, source lines 10827–10827. Cited at proof line 2149: the glued surjection.

Source reference. Original lemma 05VY: sources/stacks-chow/tags.txt, source lines 7567–7567. Cited at proof line 2153: Q flat over T.

Source reference. Original lemma 060V: sources/stacks-chow/tags.txt, source lines 7739–7739. Cited at proof line 2159: Q of finite presentation.

Lemma 0.2. *In Situation 082M let $\{X_i \to X\}_{i\in I}$ be an fpqc covering and for each $i, j \in I$ let $\{X_{ijk} \to X_i \times_X X_j\}$ be an fpqc covering. Denote $\mathcal{F}_i$, resp. $\mathcal{F}_{ijk}$ the pullback of $\mathcal{F}$ to X_i, resp. X_{ijk}. For every scheme T over B the diagram*

$$Q_{\mathcal{F}/X/B}(T) \longrightarrow \prod_i Q_{\mathcal{F}_i/X_i/B}(T) \xrightarrow[pr_1^*]{pr_0^*} \prod_{i,j,k} Q_{\mathcal{F}_{ijk}/X_{ijk}/B}(T)$$

presents the first arrow as the equalizer of the other two. The same is true for the functor $Q^{fp}_{\mathcal{F}/X/B}$.

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Proof. Let $\mathcal{F}_{i,T} \to \mathcal{Q}_i$ be an element in the equalizer of pr_0^* and pr_1^*. By Remark 08IU we obtain a surjection $\mathcal{F}_T \to \mathcal{Q}$ of quasi-coherent $\mathcal{O}_{X_T}$-modules whose restriction to $X_{i,T}$ recovers $\mathcal{F}_i \to \mathcal{Q}_i$. By Morphisms of Spaces, Lemma 05VY we see that $\mathcal{Q}$ is flat over T as desired. In the case of the functor $\mathrm{Q}^{fp}_{\mathcal{F}/X/B}$, i.e., if $\mathcal{Q}_i$ is of finite presentation, then $\mathcal{Q}$ is of finite presentation too by Descent on Spaces, Lemma 060V. $\square$